VOIDSCOPE · TRACEABILITY MATRIX

VoidScope traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/voidscope		
Source	docs/03b-traceability.md @ 463eaf8		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-02	Amish Chadha	Generated by .kit/trace.py
0.2	2026-11-02	Amish Chadha	Generated by .kit/trace.py
0.1	2026-11-02	Amish Chadha	Generated by .kit/trace.py

VoidScope traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 19 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Passes a core hole	VDS-CAL-001, VDS-BLD-001	Analysis (VDS-CAL-001 A1: 48 mm at the head, 47 mm at the grip); pass the assembled probe through a 51 mm test ring at TRL 4	Met on paper
UR-002	PR-002	Reach	VDS-CAL-001, VDS-BLD-001, VDS-STM-001	Analysis (A4g: 4,203 mm; A4: 4,131 mm before the steering control reaches the hole). Beyond about 2 m in the void the tip must rest on debris: unsupported, the tip droops 39 mm at 2 m and the factor on yield at a button hole falls to 1.4 at 4.1 m (VDS-CAL-001 B); measure at TRL 4	Met on paper
UR-003	PR-003	Vision in dark and dust	VDS-CAL-001, VDS-BLD-001	Analysis (C2: 19.2 px; C6: 7 lx after 2 m of dusty air); dust test box at TRL 4	Met on paper
UR-004	PR-004	Rubble handling	VDS-CAL-001, VDS-BLD-001	Design (IP68 camera in an aluminium tip housing, G2: 2.85 mm minimum wall); drop and immersion tests at TRL 4, with the tip straight and bent	Met by design
UR-005	PR-005	Water ceiling	VDS-CAL-001, VDS-BLD-001	Analysis (D1, D2: 1.2 mL/min at 1 m and 20 C, 4.4 mL/min at 2 m and 40 C with the bore 5 per cent oversize); timed collection at TRL 4	Met on paper
UR-005	PR-006	No flow unless bitten	VDS-CAL-001, VDS-BLD-001	Design (D7: 19.6 kPa at 2 m); valve cracking pressure and drill with the medical lead at TRL 4	To verify at TRL 4
UR-006	PR-007	Air flow and pressure	VDS-CAL-001, VDS-BLD-001, VDS-STM-001	Analysis (E3: 44 L/min at full speed; E5: 0.9 kPa at shut-off, to confirm: the blower runs from the converter's regulated output with a fixed speed cap, fix F5); flow meter and manometer at TRL 4	To verify at TRL 4
UR-007	PR-008	Battery life	VDS-CAL-001, VDS-BLD-001, VDS-STM-001	Analysis (F4: 6.2 h at 20 C, 4.35 h at -10 C); run-down tests at TRL 4	Met on paper
UR-008, UR-015	PR-009	Cost	VDS-CAL-001	Analysis (bom/bom.csv: USD 1,780; I2, I4)	Met on paper
UR-009	PR-010	Setup time	VDS-CAL-001, VDS-BLD-001	Estimate (H6: 2.2 min); timed drills with a USAR partner at TRL 4	To verify at TRL 4
UR-010	PR-011	Water path change	VDS-CAL-001, VDS-BLD-001	Design (D8: about 4 min, single-use water set); timed replacement at TRL 4	Met by design

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-011	PR-012	Look round a 45-degree bend	VDS-CAL-001, VDS-BLD-001	Analysis (J3: 90 degrees each way; J6: 83 mm to the side; C2: 19 px); test box at TRL 4	Met on paper
UR-012	PR-013	Two-way voice	VDS-CAL-001, VDS-BLD-001	Design (tip speaker used as speaker and microphone, push-to-talk headset); speech test in a rubble box at TRL 4	Met by design
UR-013	PR-014	Tip surface temperature	VDS-CAL-001, VDS-BLD-001	Analysis (K5: 44.4 C at 35 C, 53.9 C at 45 C, STM S14, fix F1); thermocouple on the tip housing at TRL 4	Met on paper
UR-014	PR-015	Commodity parts and hand tools	VDS-CAL-001	Design (build plan VDS-BLD-001, outside lathe allowance in BOM line 29); build at TRL 4	Met by design
UR-009	PR-016	Carried load	VDS-CAL-001	Analysis (B5: probe 4.18 kg; I1: surface unit 6.2 kg); weigh at TRL 4	Met on paper
UR-004, UR-002	PR-017	Rod structure	VDS-CAL-001, VDS-STM-001	Analysis (B12: 940 N; B11: 2.33; STM S1, S3); load test at TRL 4	Met on paper
UR-005, UR-013	PR-018	Steering and battery stops	VDS-CAL-001	Analysis (J16: 3.0, wire breaking load assumed 250 N); supplier data and stops S2 and S8 in the build plan	Met on paper
UR-016	PR-019	Status statement	VDS-CAL-001	Inspection of the documents	Met by design

Gaps

None.